



MINISTRY OF SCIENCE
AND HIGHER EDUCATION
OF THE RUSSIAN FEDERATION
ITMO University

49 Kronverksky ave., bldg. A
St. Petersburg, Russia, 197101

Dear Admissions Committee,

I am writing to recommend Mr. Ayan Ali for admission to your PhD program. I am Dr. Aleksey Vedyakov, Associate Professor and Vice Dean at the Faculty of Control Systems and Robotics, ITMO University, and a researcher at the International Laboratory "Geometric Methods of Control and Applications" and the International Research Center "Nonlinear and Adaptive Control Systems."

I taught Ayan in the courses "Machine Learning in Robotics I" and "Machine Learning in Robotics II" during his master's program in Robotics and Artificial Intelligence. These courses require a solid command of both theoretical foundations and practical implementation of data-driven algorithms for robotic perception, decision-making, and control. Ayan achieved the highest grade (5) in both courses, a result attained by only a small fraction of students in the program.

Throughout the coursework, Ayan consistently demonstrated strong analytical thinking, good mathematical maturity, and a disciplined approach to his work. He showed a clear understanding of machine learning fundamentals, including supervised and reinforcement learning, and was able to apply these methods to problems in robotic motion and control. He often went beyond the basic course requirements by refining model architectures, experimenting with real-time learning setups, and exploring hybrid AI-control approaches.

Ayan's performance in laboratory and project work was consistently strong. He developed well-structured and reproducible code and demonstrated a sound understanding of algorithmic design principles. His ability to reason about the integration of learning algorithms into control pipelines, including adaptive and model-predictive frameworks, corresponds to the level typically expected from early-stage researchers.

In addition to his academic performance, Ayan is a motivated and collaborative student. He actively participates in technical discussions, works well in group settings, and approaches research problems in a professional and responsible manner. These qualities make him well suited for both academic research and applied development work.

In my professional opinion, Ayan Ali possesses the background, motivation, and persistence required for advanced research in robotics, artificial intelligence, and control systems. I consider him a strong candidate for PhD studies and recommend him for admission to your program.

If you require further information, please contact me at vedyakov@itmo.ru.

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